

The phonetics of Japanese maid voice I: A preliminary study

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Background

- Meido kissa developed in the last ten years or so, mainly in Akihabara. (01/03/30; Takatora 2012).
- Maids working in meido kissa prototypically use specific phrases.   
- They also use a “distinctive tone of voice”.
- (Not every maid changes their voice. And there are many kinds of meido kissa, and accordingly many kinds of maids.)
- Question: what do they exactly do when they use maid voice? What's the acoustic characteristics of maid voice?

Bigger questions

- What is *moe* (from a linguistic perspective)?
- Even more generally, what kinds of voice do listeners find attractive? (assuming that *moe* voice is attractive).
- To the extent that maids are aiming for “cuteness” in using maid voice, what do they do to make them sound cute?

Method

Method: Recording place

- Maid riture, Fèlicie at Akihabara.
- <http://www.felicie-web.com/>

Method: Participants

Maid R

Maid M

Maid S

Who they are and what they did

- Maid R
2 years at meido rifure; 3 years at meido kissa.
- Maid M
6 years at meido rifure (not continuously).
- Maid S
5 years at meido rifure.
- The task: to read phrases and sentences in normal voice and maid voice.

Method: Stimuli

- A warm-up: some typical maid phrases.
- Four sentences with sonorants to measure intonational contours: 2 rep*(4 norm => 4 maid).
- A set of five Japanese vowels (to measure formants), in isolation and preceded by different types of consonants: 2 rep * (10 norm => 10 maids) * 3 orders.

Method: Recording settings

- DR 40 (Tascam).
- 44.1k sampling frequency and 16 bit of quantization level.
- Recording took place in a quiet room at Fèlicie in Akihabara.
- The data from Maid M is set aside, because of her quiet voice and a recording problem.

Method: Acoustic values measured

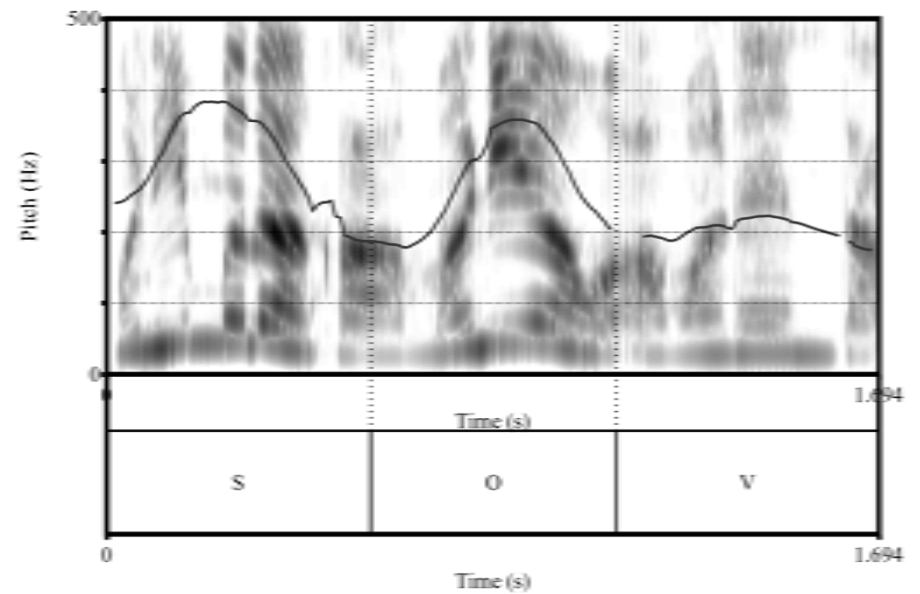
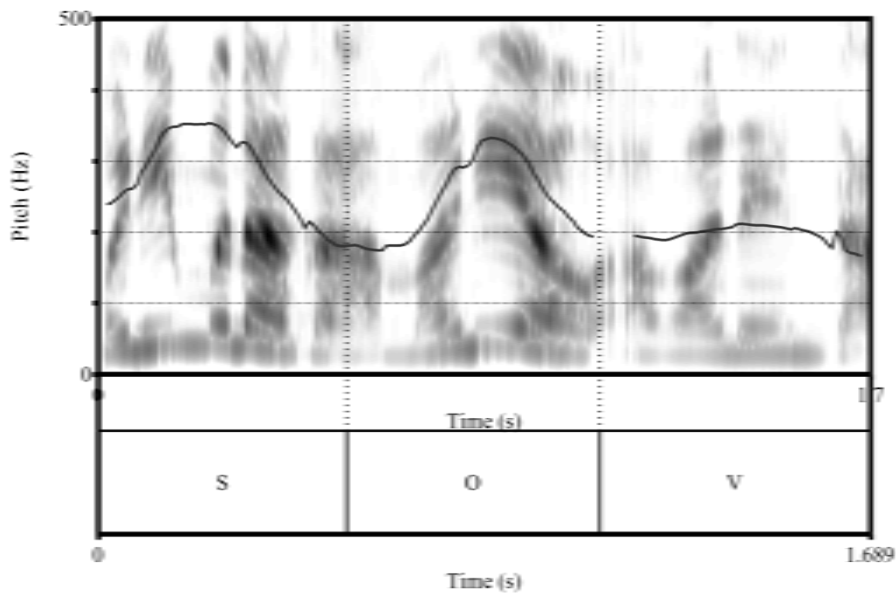
- A bit shot in the dark.
- F0 (fundamental frequency): higher sounds may sound cuter.
- Other vocalic, acoustic values.
- This study is really exploratory, so the analysis proceeded “as we found things”.

Results

Intonation

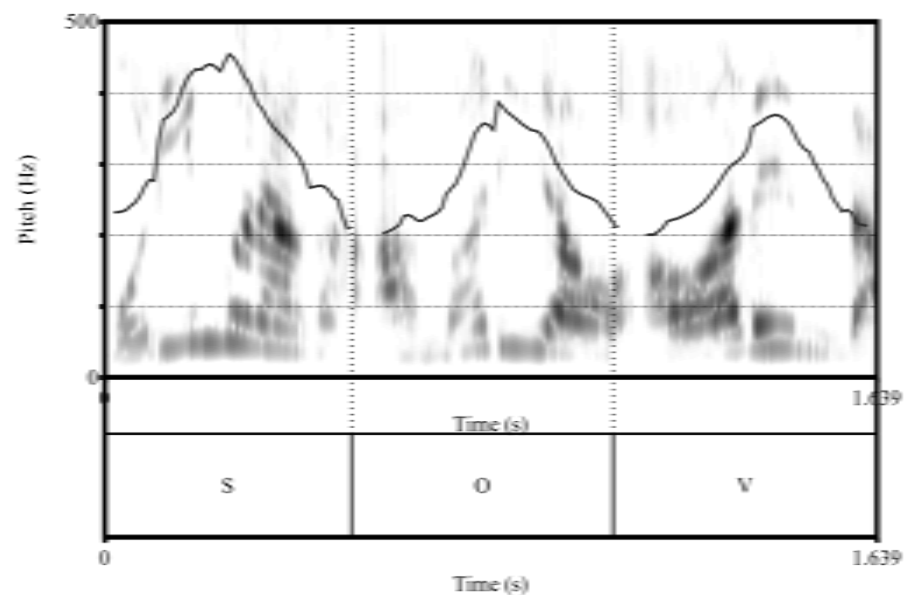
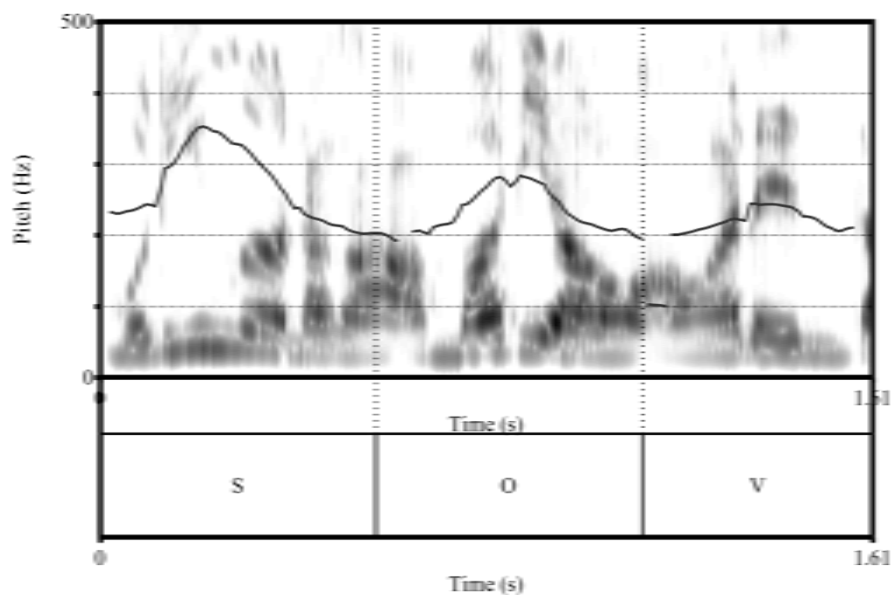
Intonational contour: Maid R

- 森村がアマリアを哀れんだ。

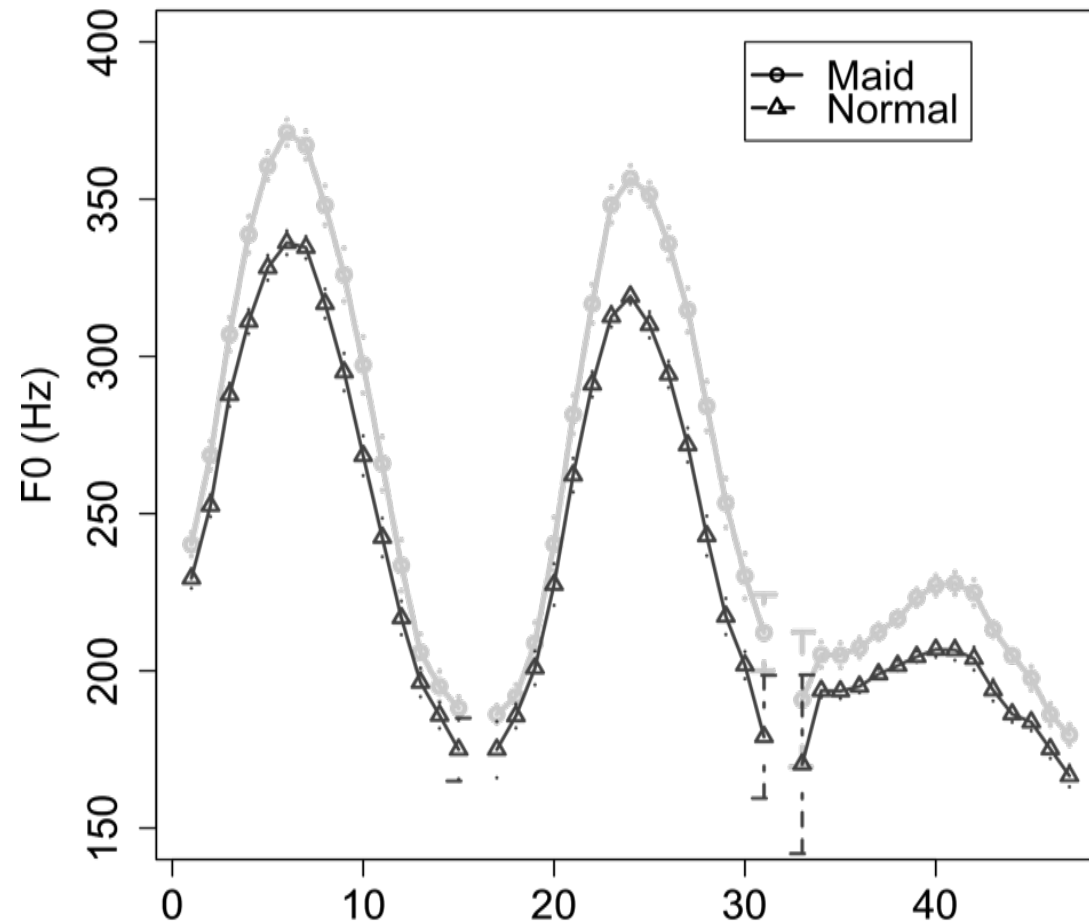


Intonational contour: Maid S

- 森村がアマリアを哀れんだ。

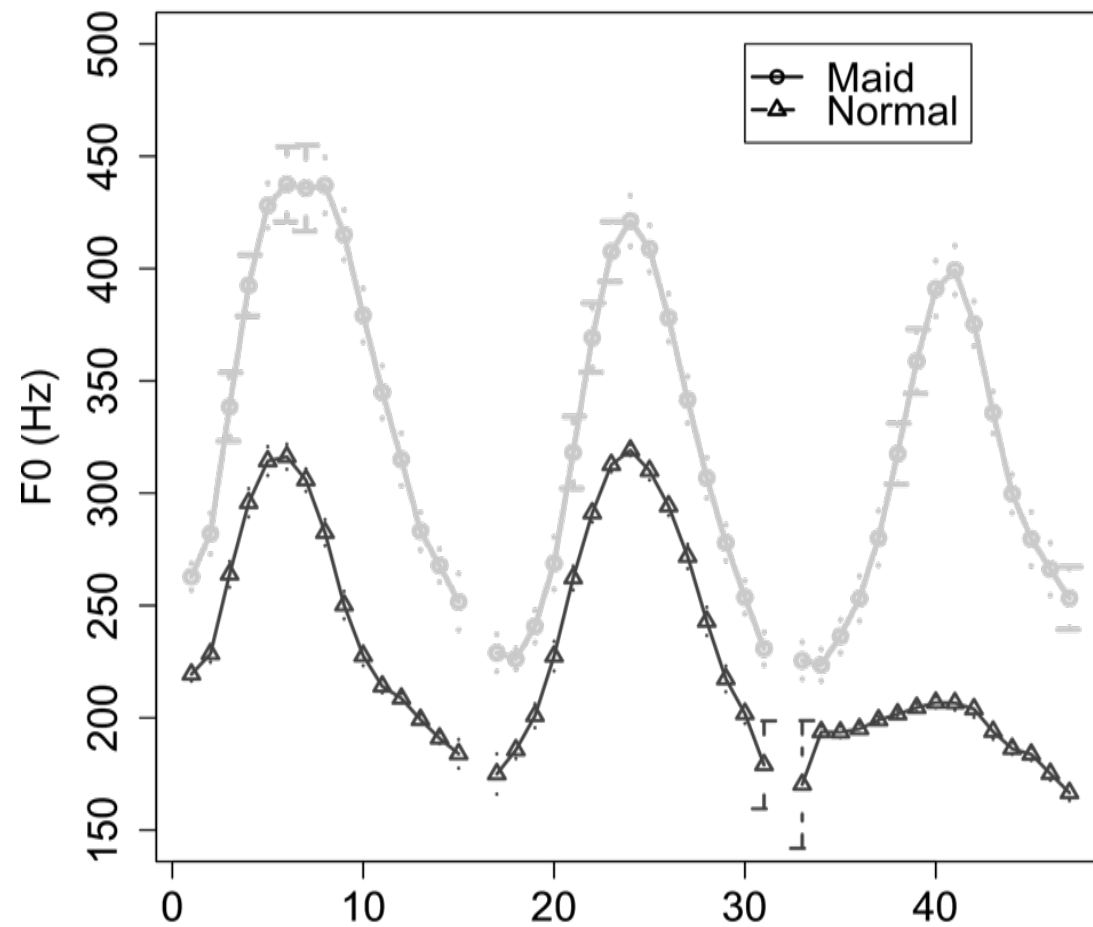


Intonational contour: Maid R



Schematized F0 contour of SOV. Each phrase was divided into 15 equally-timed intervals. Averaged over 4 lexical sets and 8 repetitions.

Intonational contour: Maid S



- They both show generally higher pitch in maid voice.
- Maid R: H targets are clearly raised; L targets are slightly raised (?).
- Maid S: Both L and H targets are raised, but H targets are raised more than L targets.

Stats: Maid R

- Linear mixed model with the voice type (maid vs. normal) and the syntactic condition as fixed factors, lexical sets as a random factor.
- The effects of the voice type:
 - L: $t=-1.812$, *n.s.*
 - H: $t=-16.23$, $p<.001$
 - Rise: $t=-2.97$, $p<.01$
- H targets are raised. No statistical confirmation about the raising of L-targets.

Stats: Maid S

- The effects of voice types for Maid S:
 - L: $t=-6.71$, $p<.001$
 - H: $t=-26.91$, $p<.001$
 - Rise: $t=-16.69$, $p<.001$
- Both L and H targets are raised.
- However, the significant difference in the magnitude of rise indicates that H is raised more.

Speech rate

- Do the maids speak slower in their maid voice?

Maid R	SUBJECT	OBJECT
Norm	56.8 (1.0)	55.1 (1.7)
Maid	56.4 (0.8)	55.2 (1.4)
Maid S	SUBJECT	OBJECT
Norm	56.2 (1.07)	55.8 (1.3)
Maid	53.6 (1.0)	55.9 (1.2)

Duration of each interval (ms) with margins of error.

Change in F0/msec

Dynamic F0 movement at the same speech rate implies greater F0 changes/(m)sec:

Maid R	SUB-Rise	SUB-Fall	OBJ-Rise	OBJ-Fall
Norm	5.1 (0.17)	-4.0 (0.14)	6.2 (0.41)	-6.4 (0.54)
Maid	6.3 (0.17)	-4.8 (0.14)	6.9 (0.35)	-7.6 (0.37)
Maid S	SUB-Rise	SUB-Fall	OBJ-Rise	OBJ-Fall
Norm	5.9 (0.35)	-3.3 (0.19)	4.3 (0.54)	-2.8 (0.58)
Maid	9.9 (0.79)	-8.9 (0.84)	9.6 (0.66)	-8.9 (0.47)

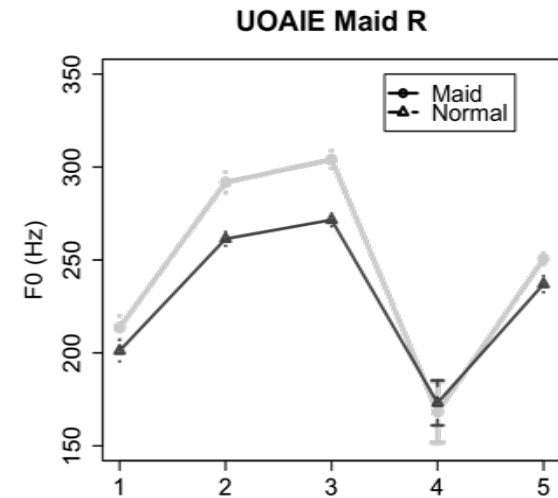
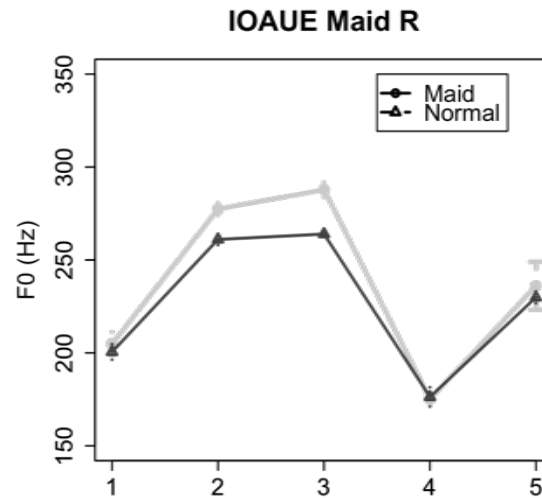
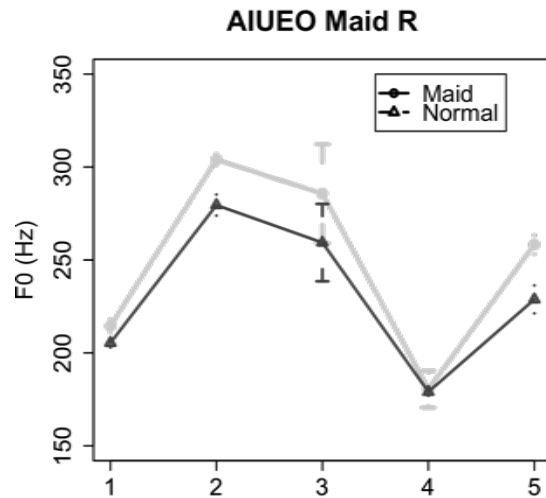
F0 change rate (F0/msec) for rise and fall in subject and object.

Conclusion about F0

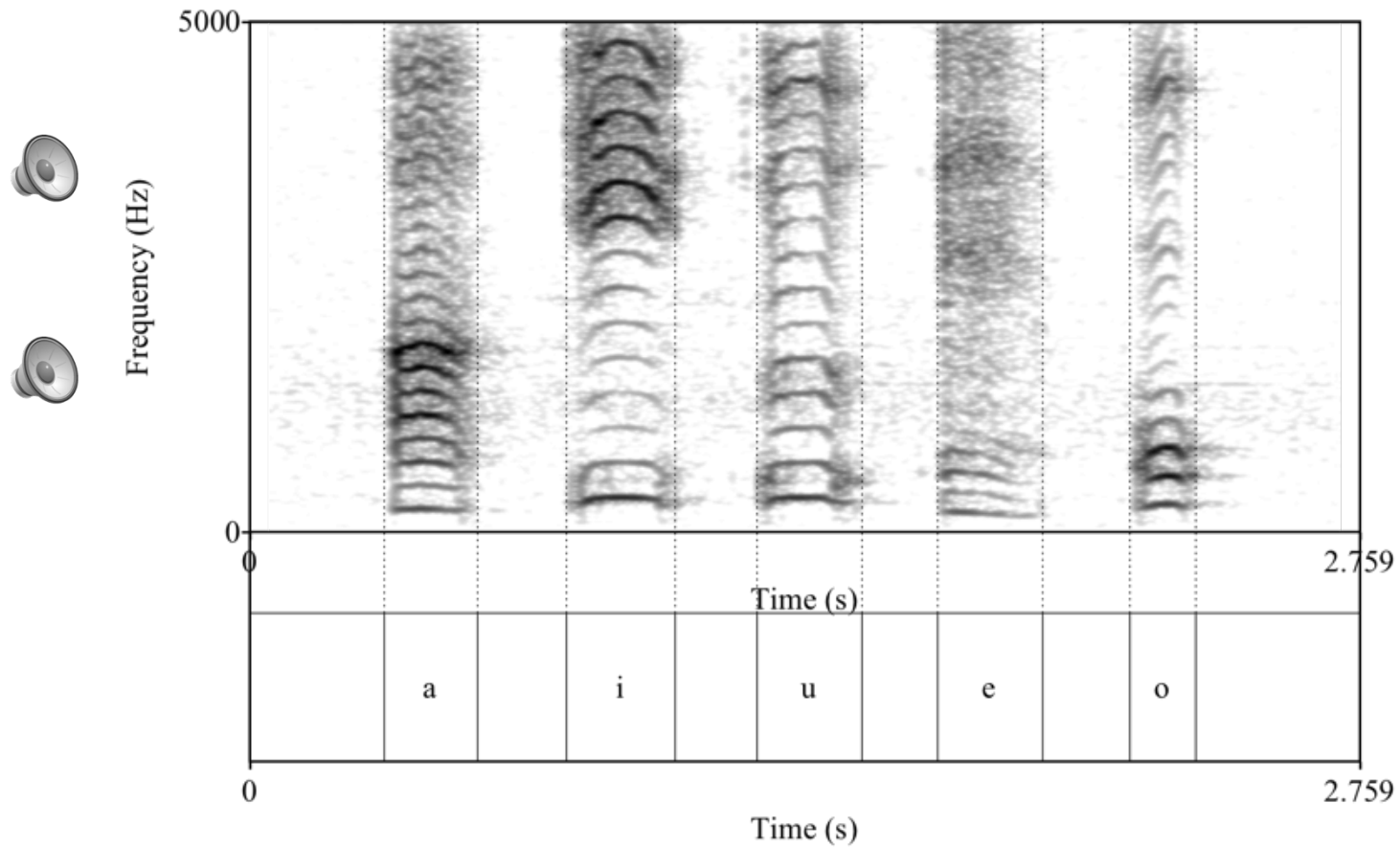
- Both maids raised their pitch.
- Maid R raised only H targets.
- Maid S raised both targets, but raised H targets more extensively.
- (Conforming to Fujisaki 1983 model?)
- Neither maids changed speech rate; instead they changed f0 more dynamically.

Some observations about
the AIUEO contour

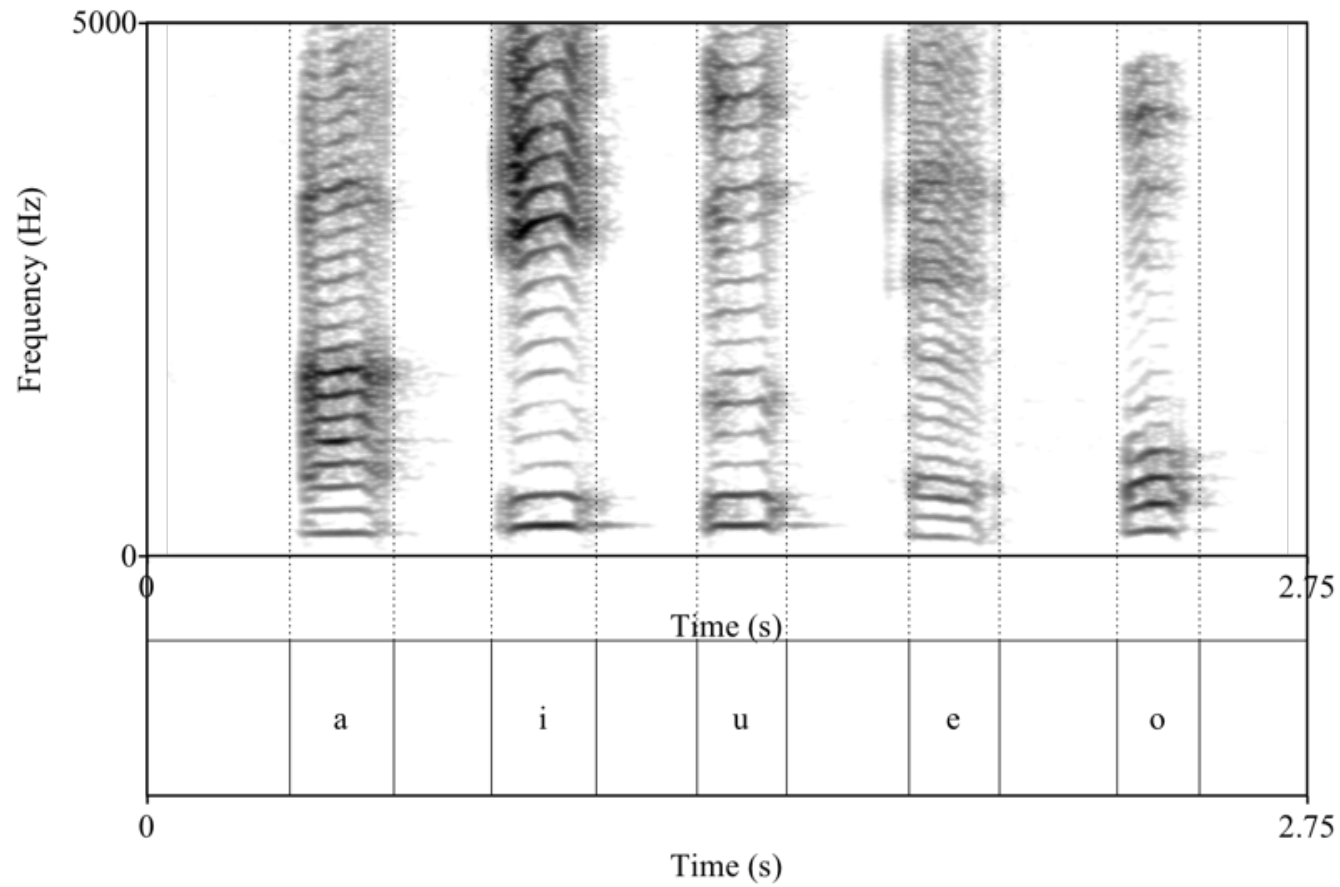
AIUEO reading F0 contour: Maid R



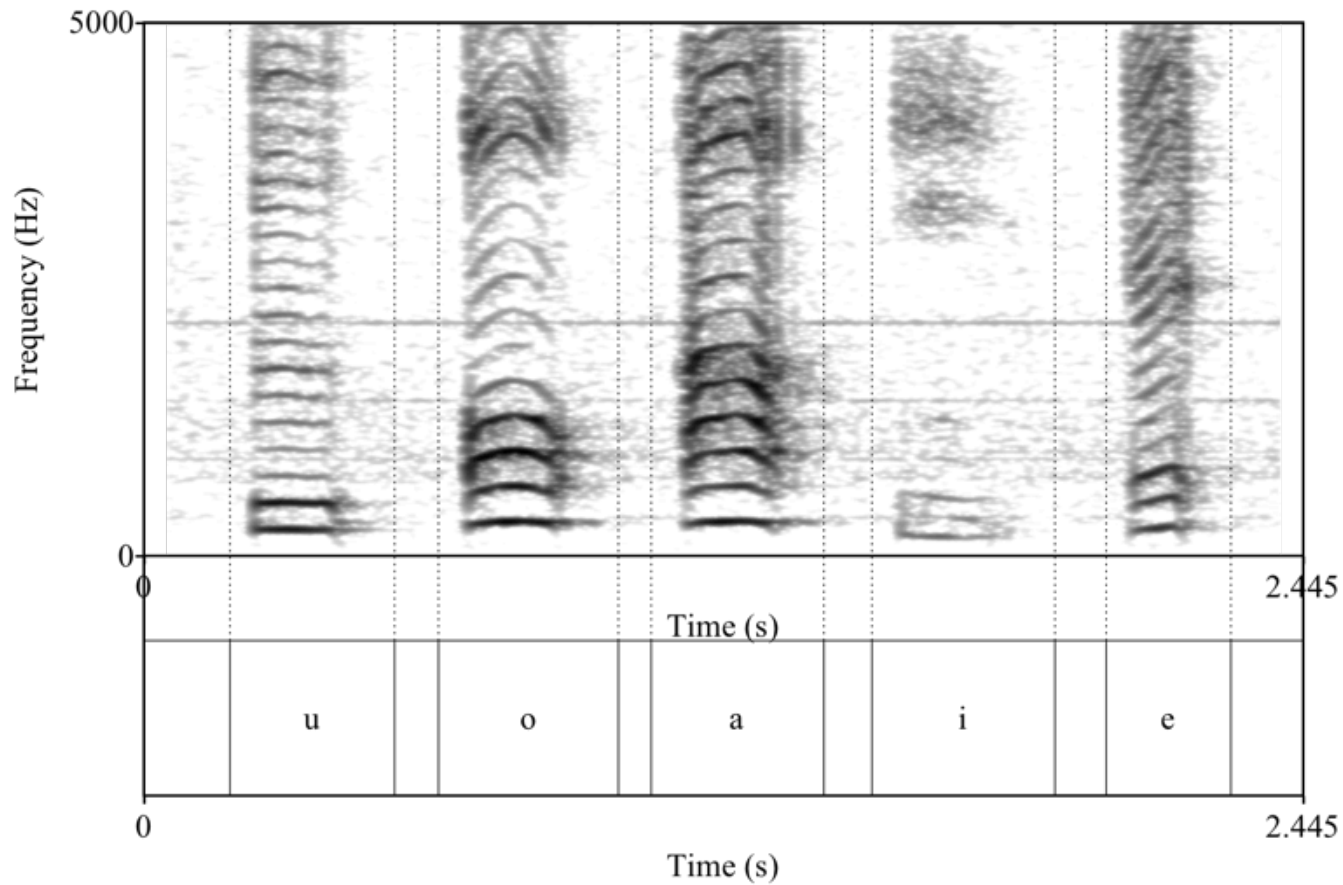
Penultimate breathiness: Maid R



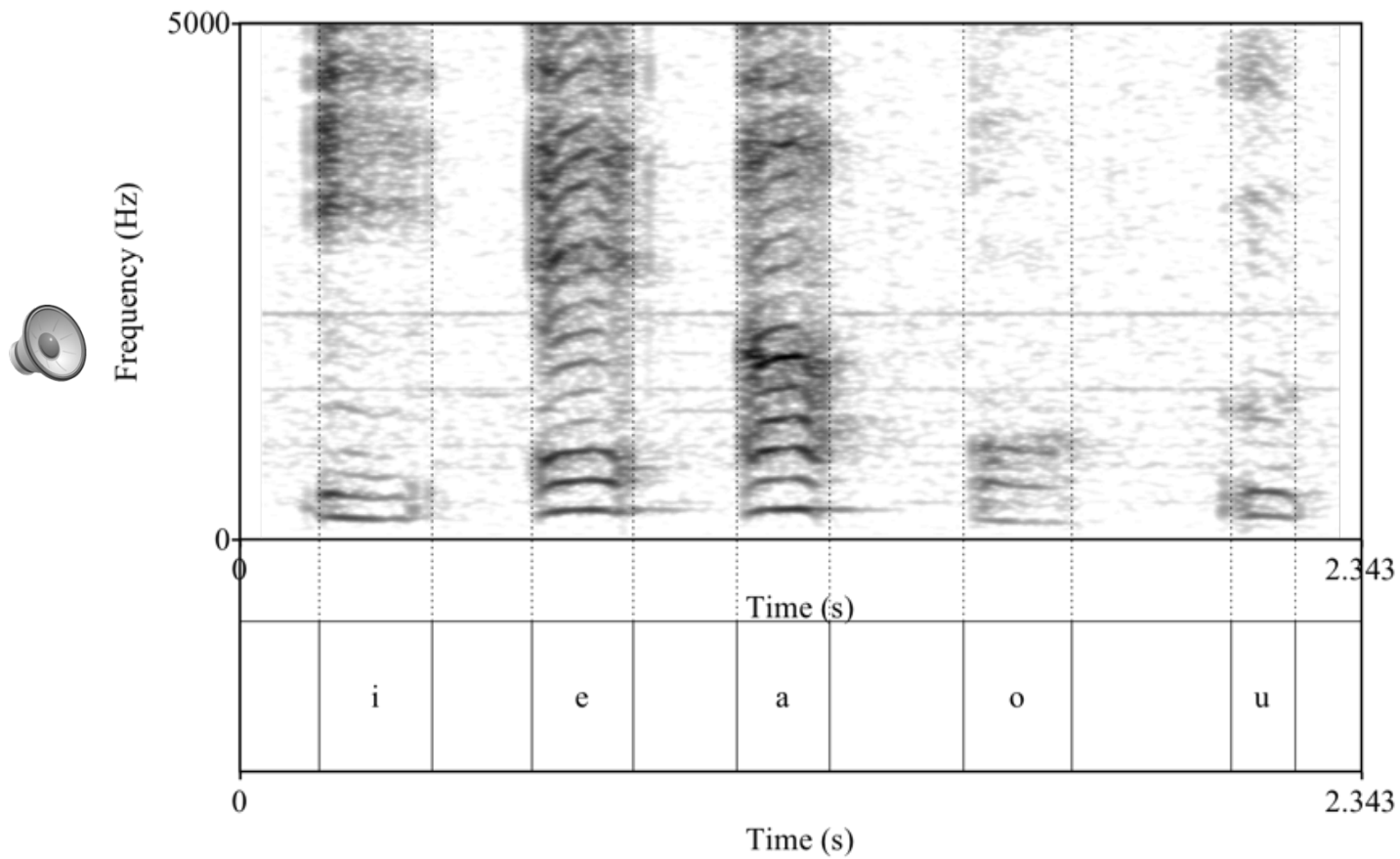
No breathiness in normal speech



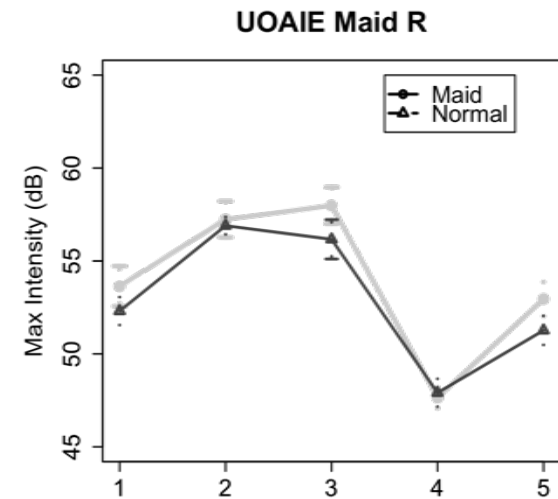
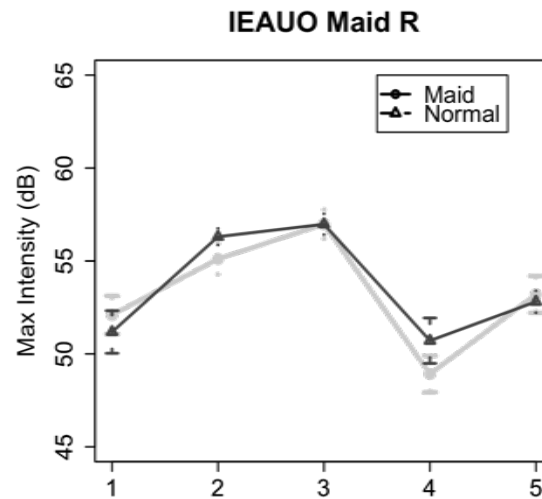
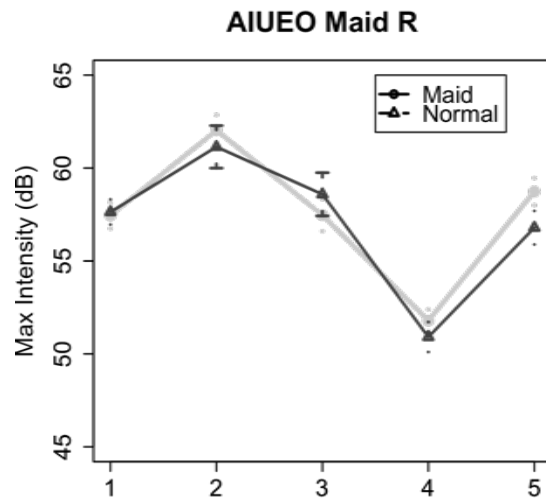
Is it positional or inherent to [e]?



Is it positional or inherent to [e]?



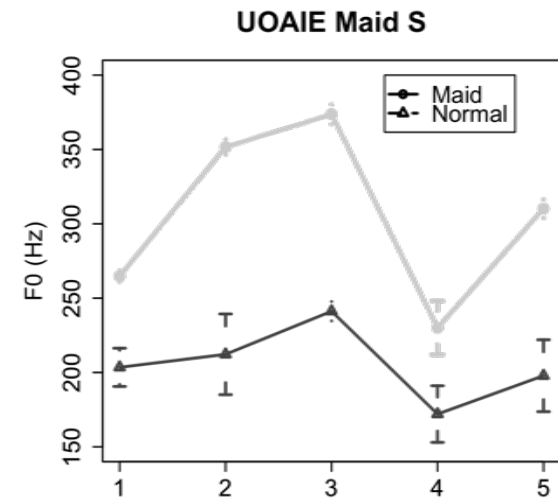
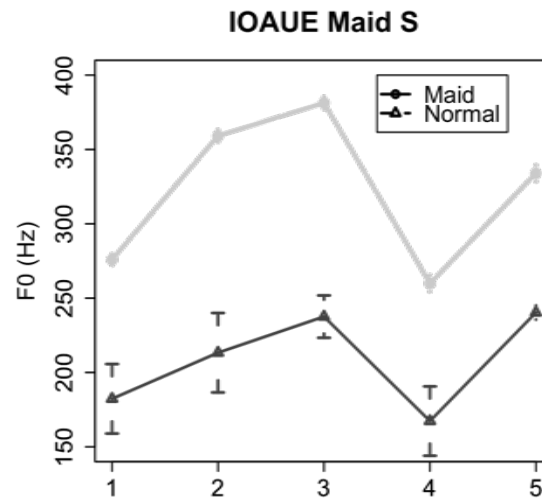
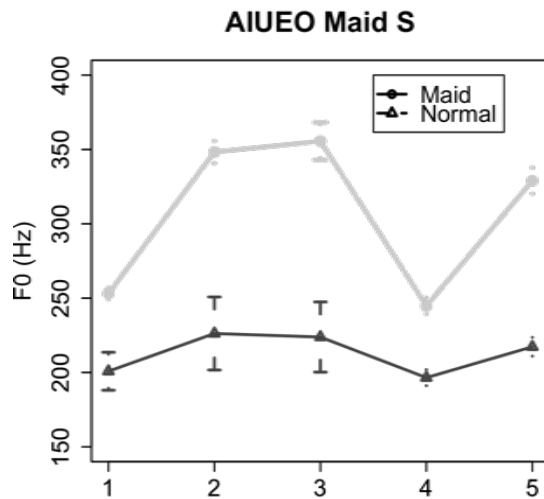
Intensity contour



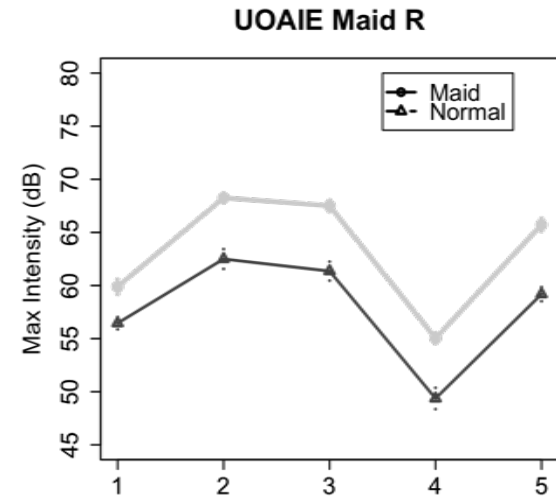
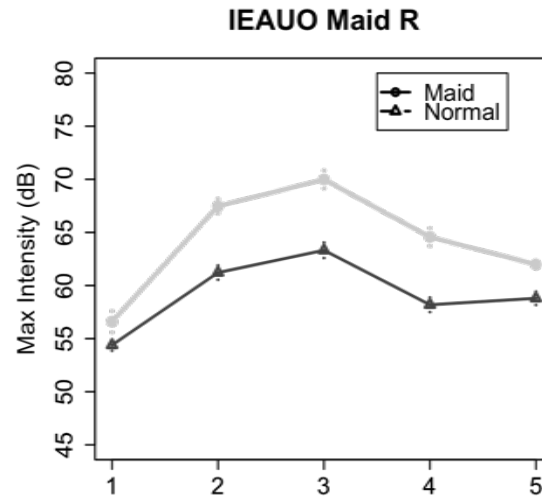
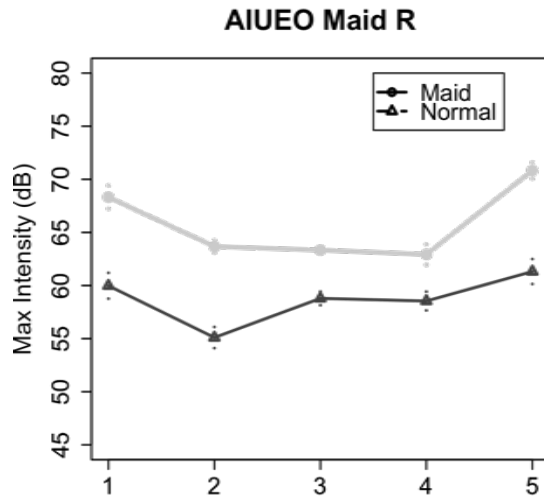
Summary: Maid R

- Dynamic F0 contour.
- Penultimate breathiness.
- No drastic changes in the intensity contour.

Now moving onto Maid S: F0 contour



Intensity contour: Maid S



Summary: Maid S

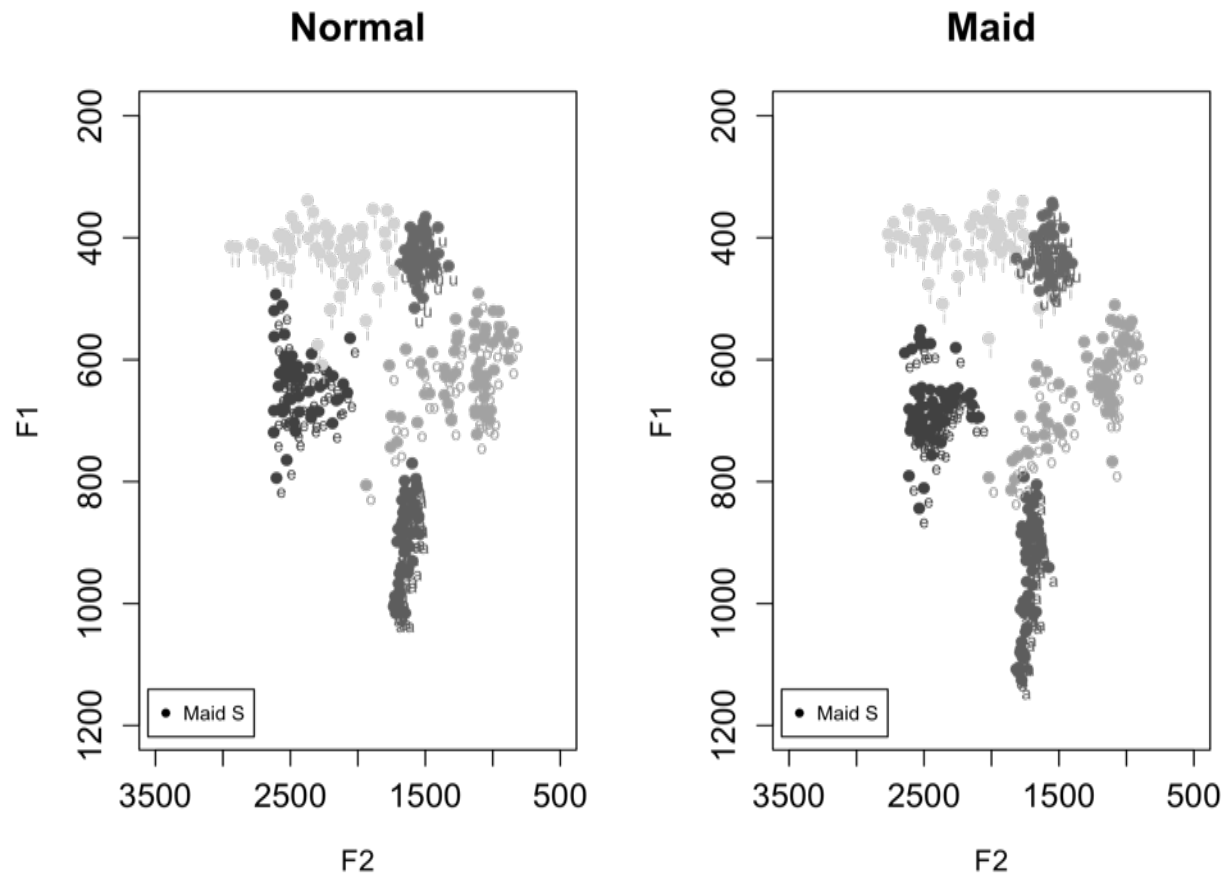
- No breathiness.
- Clear differences in F0 and intensity contours.

Formant analysis

Formant analysis

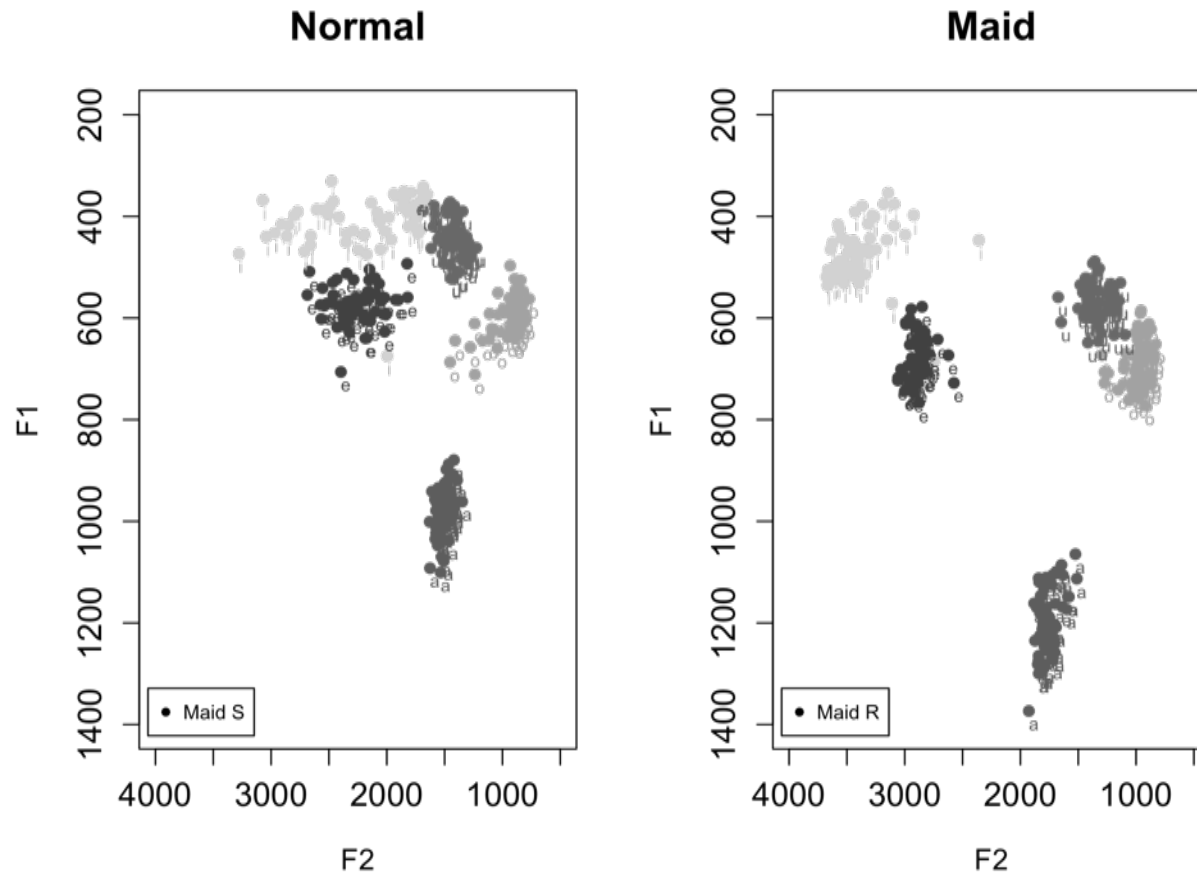
- 5 formants within 6000Hz for Maid R.
- 5 formants within 6000Hz for normal voice and 7000Hz for maid voice for Maid S.

Formant analyses: Maid R



- [i] seems characteristically less fronter in maid voice.
- F2 average: norm=2264Hz; maid=2110Hz.
- My guess: reduces the images of sharpness.
- [a] seems lower in maid voice
- F1 average: norm 900Hz; maid=959Hz.

Vowel Space: Maid S



- [i] is much fronter in maid voice.
- F2 average: norm=2170Hz; maid=3340Hz.
- [e] is also much fronter in maid voice.
- F2 average: norm=2261Hz; maid=2888Hz.

- [a] seems lower in maid voice.
- F1 average: norm=978Hz; maid=1199Hz.
- [u] is lower and backer in maid voice.
- F1 average: norm=443Hz; maid 565Hz.
- F2 average: norm=1425Hz; maid 1338Hz

Characteristics of maid formants

- Both maids show larger opening of the mouth for [a].
- Maid R shows backer [i] whereas Maid S shows fronter [i, e].
- My impressionistic guess (unsubstantiated):
Maid R gives a less sharp impression, whereas Maid S is shooting for a more lively character.

Other values tested...

- Values that are used in the speech pathology studies related to voice quality.
 - Harmonic-to-noise ratio
 - Jitter
 - Simmer
- Unfortunately no consistent patterns.

An addendum

- What do they think they are doing when they make maid voice?
- 「地声の上澄みをとっている」(Maid R, p.c. 1/28/2012)
- Emphasizing high frequency components?
- A functional motivation according to her: useful as she needs to distribute fliers.

Conclusion

Conclusions

- Maids do (or can) change their voice.
- Two different maids take different strategies, although they do share some strategies.
- Maid R:
 - Raising of F0 H targets.
 - More dynamic F0 movement per msec.
 - Penultimate breathiness.
 - Backing of [i] and lowering of [a].
- Maid S:
 - Raising of F0 L and H targets (H-raising more extensive).
 - More dynamic F0 movement per msec.
 - Fronting of [e, i] and lowering of [a].

What's coming next

- More acoustic analyses. Ideally more maids.
- I have recorded two voice actresses producing materials in normal voice, tsun maid voice and moe maid voice.
- Just to illustrate, the first two sentences of 北風と太陽.



“The usual slide” that often appears at the end

- Instead of me thanking you, let’s let them do it...



- And of course...



To be continued to II (which
will probably be about
voice actresses)...

Acknowledgements

- Fèlicie (especially the tenchoo-san) who allowed me to record the maids.
- The three maids for participating in the production experiment, and also their help with these slides.
- Prof. Kazuko Shinohara for her continued support on this and related projects.
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